



FPT UNIVERSITY

Faculty of Artificial Intelligence

Capstone Project Report

**Personalized and Edge-Efficient Vietnamese
Compact Writing Restoration with
Keyboard-Aware Typo Correction**

NextKey		
Group Members	Student Name 1	Student ID 1
	Student Name 2	Student ID 2
	Student Name 3	Student ID 3
Supervisor	Title & Supervisor Name	
Ext Supervisor	Title & External Supervisor Name (Optional)	
Capstone Project Code	AIP491	

Da Nang, August 2026

Abstract

Rapid digital communication on mobile devices has led to widespread adoption of compact, informal Vietnamese writing characterized by unaccented text, omitted inter-word whitespace, and keyboard-adjacent typing errors on virtual touchscreens. Traditional rule-based input method editors (Input Method Editor (IME)) and statistical models are incapable of rectifying such multi-faceted compound corruptions, while large language models are computationally prohibitive for client-side edge deployment.

This capstone project presents **NextKey**, an edge-efficient artificial intelligence system engineered to simultaneously perform character typo correction, word boundary segmentation, and diacritic restoration in a single inference pass. We introduce the **JDWR v1** multi-domain dataset ($> 840,000$ sentences) and a realistic keyboard-adjacency noise engine simulating mobile **QWERTY! (QWERTY!)** typing behaviors. We propose **CascadeTriBiGRU**, a novel architecture incorporating local 1D convolutions, a shared bidirectional Gated Recurrent Unit (GRU) backbone, and hierarchical cross-head feature gating to condition diacritic assignment on corrected characters and word boundaries.

Extensive empirical evaluations across 4 research phases demonstrate that **CascadeTriBiGRU** achieves a State-of-the-Art clean Character Error Rate (Character Error Rate (CER)) of 4.79%, Word Accuracy of 80.44%, Typo Recovery Rate of 65.92%, and a Sentence Near-Perfect Rate ($\text{CER} \leq 5\%$) of 59.57% on the full held-out test suite. Through Knowledge Distillation and 8-bit Signed Integer (INT8) quantization, the model is compressed to under 60 KB with a sub-millisecond latency (0.25 ms on Central Processing Unit (CPU)), delivering a robust, privacy-preserving, offline-capable restoration engine for next-generation intelligent mobile keyboards.

Keywords: Vietnamese Natural Language Processing, Compact Writing Restoration, Diacritic Restoration, Keyboard Typo Correction, Multi-Task Learning, CascadeTriBiGRU, Edge AI, Model Quantization, Knowledge Distillation.

Acknowledgments

This section is used for you to express your gratitude to those who help you to finish the capstone.

Contents

Abstract	1
Acknowledgments	2
Definition and Acronyms	7
I. Project Introduction	9
1. Overview	9
1.1. Project Information	9
1.2. Project Overview	9
2. Project Background	10
2.1. Overview of the Field	10
2.2. Existing Systems and Related Work	11
2.3. Limitations of Existing Solutions	11
3. Problem Statement	12
4. Research Questions	12
5. Project Objectives	13
6. Significance of the Project	14
7. Project Scope & Limitations	14
7.1. In-Scope	14
7.2. Out-of-Scope & Limitations	14
II. Project Management Plan	16
1. Team Work	16
1.1. Team Structure and Roles	16
1.2. Communication Plan	17
2. Project Management Approach	17
2.1. Work Breakdown Structure	17
2.2. Risk Management	18
2.3. Quality Management Plan	19
2.4. Budget and Resource Allocation	19
2.5. Change Management Process	20

2.6.	Project Closure and Deliverables	20
III.	Methodology	21
1.	Overview of Framework/Approach	21
2.	Data Collection and Preprocessing	21
3.	Feature Selection and Engineering	21
4.	Proposed Model/System/Method (Contributions)	21
5.	Experimental Settings	21
6.	Evaluation Metrics	22
IV.	Results and Discussion	23
1.	Results and Analysis	23
2.	Discussion	23
3.	Personal Reflection (Optional)	23
V.	System Design and Implementation	24
1.	AI Model Integration	24
2.	Data Flow and Processing	24
3.	Deployment Strategy	24
4.	Scalability and Maintenance	24
VI.	Conclusion	25
1.	Summary of Findings	25
2.	Contributions and Reflections on the Project	25
3.	Limitations and Future Work	25
VII.	Ethical Declaration	26
1.	Ethical Considerations	26
2.	Declaration of AI-based Assistance	26
VIII.	Appendices	27
1.	First Appendix Sample	27
2.	Second Appendix Sample	27
IX.	References	29

List of Figures

List of Tables

II..1	Team Structure, Research Tracks, and Role Allocations	16
II..2	Work Breakdown Structure (WBS) and Research Project Timeline	18
II..3	Project Risk Register and Mitigation Framework	19

Definition and Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BF1	Boundary F1-Score
BiGRU	Bidirectional Gated Recurrent Unit
BiLSTM	Bidirectional Long Short-Term Memory
BLEU	Bilingual Evaluation Understudy
CER	Character Error Rate
CLI	Command Line Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CRF	Conditional Random Fields
DL	Deep Learning
EM	Exact Match
FP32	32-bit Floating Point
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
IME	Input Method Editor
INT8	8-bit Signed Integer
JDWR	Joint Diacritic and Word Restoration
KD	Knowledge Distillation
LLM	Large Language Model
LSTM	Long Short-Term Memory

ML	Machine Learning
NLP	Natural Language Processing
ONNX	Open Neural Network Exchange
OOD	Out-of-Distribution
PTQ	Post-Training Quantization
QKD	Quantization-Aware Knowledge Distillation
RAM	Random Access Memory
REST	Representational State Transfer
RNN	Recurrent Neural Network
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SOTA	State-of-the-Art
TCN	Temporal Convolutional Network
UI	User Interface
WAcc	Word Accuracy
WBS	Work Breakdown Structure
WER	Word Error Rate

I. Project Introduction

1. Overview

1.1. Project Information

- **Project Title:** NextKey: Personalized and Edge-Efficient Vietnamese Compact Writing Restoration with Keyboard-Aware Typo Correction
- **Short Title:** NextKey: Vietnamese Compact Writing Restoration
- **Project Code:** AIP491 (Capstone Project in Artificial Intelligence)
- **Academic Institution:** Faculty of Artificial Intelligence, FPT University Da Nang Campus
- **Team Members:**
 - Member 1: AI / Deep Learning Research Specialist (Student ID: DE18xxxx)
 - Member 2: Edge Systems & Model Optimization Engineer (Student ID: DE18xxxx)
 - Member 3: Data Engineering & Full-Stack Lead (Student ID: DE18xxxx)
- **Supervisor:** Academic Supervisor, Faculty of Artificial Intelligence

1.2. Project Overview

In contemporary digital communication, text input on mobile touchscreens has become an indispensable aspect of daily productivity and social interaction. However, typing Vietnamese on compact virtual keyboards remains inherently challenging due to geometric constraints, complex tonal diacritics, and rapid typing habits. Consequently, users frequently adopt unaccented shorthand, omit inter-word whitespace, make keyboard-adjacent typing mistakes, leave residual IME composition codes, and employ informal abbreviations.

The **NextKey** project proposes an open, research-driven framework for end-to-end Vietnamese compact writing restoration. Rather than treating diacritic restoration, word segmentation, and typographical error correction as separate, disconnected tasks, NextKey formulates the problem as a unified sequence transformation task optimized for sub-millisecond edge execution.

This document is structured as follows:

-
- Chapter I. Project Introduction:** Establishes the research motivation, formal problem definition, critical assessment of existing systems, foundational research questions, objectives, significance, and operational scope.
- Chapter II. Project Management Plan:** Details the organizational structure, communication protocols, multi-stage Work Breakdown Structure (WBS), risk register, quality control framework, and change management procedures.
- Chapter III. Methodology & Architecture Exploration:** Formulates the mathematical framework, realistic synthetic noise taxonomy, candidate neural model exploration spaces, multi-task conditioning strategies, and edge compression techniques.
- Chapter IV. Results & Empirical Discussion:** Reports systematic benchmark findings across architectural families, topological scaling regimes, edge quantization paradigms, and cross-domain generalization protocols.
- Chapter V. System Design & Interactive Demonstration:** Outlines the modular backend service, client interfaces, and real-time input pipeline.
- Chapter VI. Conclusion & Ethical Considerations:** Synthesizes the core research findings, acknowledges limitations, and outlines future research trajectories.

2. Project Background

2.1. Overview of the Field

Vietnamese is an Austroasiatic, tonal language that utilizes the Latin-based *Quoc Ngu* writing system. The orthography incorporates a sophisticated system of diacritical marks: 6 tone registers (unaccented, acute, grave, hook above, tilde, and dot below) combined with phonetic vowel and consonant modifications. The omission of diacritics introduces pervasive lexical ambiguity; for instance, the unaccented syllable "ca" may correspond to multiple distinct words such as "ca" (song/case), "cá" (fish), "cà" (eggplant), or "c ấa" (all).

In mobile computing, text entry primarily relies on rule-based composition standards such as Telex and VNI. Under time-constrained or casual typing scenarios, users frequently type phonetically without completing accent key combinations, concatenate words to minimize keystrokes, and touch adjacent virtual keys inadvertently on standard **QWERTY!** layouts. Resolving these compound noisy inputs requires a synthesis of Natural Language Processing (NLP), multi-task sequence modeling, and efficient on-device edge computing.

2.2. Existing Systems and Related Work

Contemporary literature and industrial solutions can be categorized into four primary paradigms:

1. **Rule-Based Input Method Editors (IMEs):** Standard software engines such as Unikey, OpenKey, and EVKey depend strictly on deterministic keystroke composition tables. They operate at the character-entry level without contextual awareness and cannot retroactively reconstruct unaccented or typo-corrupted sentences.
2. **Commercial Smart Keyboards:** Advanced mobile keyboards such as Google Gboard, Microsoft SwiftKey, and Laban Key incorporate local n-gram language models and statistical candidate prediction. However, their corrective mechanisms operate predominantly on sliding local windows and struggle when confronted with continuous, unsegmented, multi-error streams.
3. **Statistical and Classical Diacritic Restoration Models:** Traditional research efforts have utilized n-gram language models, Hidden Markov Models (HMM), and Conditional Random Fields (CRF) [1]. These models universally presuppose that input text is already accurately tokenized into distinct words and completely devoid of spelling errors, rendering them fragile in realistic mobile environments.
4. **Pretrained Transformer and Large Language Models:** Recent approaches employ deep neural encoders and autoregressive decoders such as PhoBERT [2], ViT5, and multilingual text-to-text models (mT5, ByT5) [3]. While these architectures demonstrate strong contextual representation, their immense parameter footprint ($> 100\text{M}$ parameters) and high inference latency ($> 300\text{ ms}$) prevent direct, offline execution on resource-constrained mobile client hardware.

2.3. Limitations of Existing Solutions

A critical review of the literature reveals three fundamental technological and methodological gaps:

- **Fragmented Pipeline Assumption:** Existing systems isolate diacritic recovery, word boundary segmentation, and spell correction into sequential stages. Consequently, an error generated in an upstream module (e.g., erroneous word boundary detection) cascades uncontrollably into subsequent accentuation stages.
- **Unrealistic Noise Modeling:** Most benchmark datasets are artificially generated by merely stripping diacritics from clean corpora. They fail to capture the compound noise distributions encountered in real-world mobile input, including geometric keyboard-neighbor typos, Telex composition residue, and abbreviation colloquialisms.

-
- **Edge Execution and Privacy Infeasibility:** Cloud-reliant neural systems introduce network transmission latency, depend on continuous internet availability, and pose severe privacy vulnerabilities regarding the exfiltration of sensitive user keystroke data.

3. Problem Statement

Let an observed, corrupted input character sequence of length T be represented as:

$$X = (x_1, x_2, \dots, x_T), \quad x_t \in \mathcal{V}_{\text{input}} \quad (3..1)$$

where X may concurrently exhibit:

1. Complete absence of Vietnamese tonal markers and vowel/consonant modifications.
2. Partial or total omission of inter-word whitespace boundaries.
3. Typographical errors resulting from spatial key proximity on virtual **QWERTY!** keyboards (e.g., unintended substitution of adjacent keys).
4. Phonetic abbreviations and residual multi-stroke typing sequences (e.g., Telex keystroke artifacts).

The overarching research goal of **NextKey** is to learn a robust, parameterized mapping function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ that accurately reconstructs the clean, canonical Vietnamese sentence:

$$Y = (y_1, y_2, \dots, y_M), \quad y_m \in \mathcal{V}_{\text{output}} \quad (3..2)$$

while satisfying strict operational boundaries for client-side edge deployment: peak inference latency $\tau < 1.0$ ms on standard mobile CPU architectures and an overall model storage footprint $\mathcal{M} < 2.0$ MB.

4. Research Questions

The theoretical and empirical investigation of NextKey is structured around six core research questions:

- **RQ1 (Restoration Feasibility):** Can a unified machine learning system accurately restore standard Vietnamese text from compact input corrupted simultaneously by missing diacritics, merged spacing, abbreviations, and keyboard-adjacent typos?

-
- **RQ2 (Architectural Suitability):** Which model family (recurrent sequence taggers, temporal convolutional networks, lightweight transformer encoders, or sequence-to-sequence decoders) offers the optimal Pareto trade-off between restoration quality and edge efficiency?
 - **RQ3 (Noise Modeling Impact):** Does the incorporation of keyboard-aware synthetic noise distributions significantly improve restoration robustness on human-written compact text compared to conventional accent-stripping noise models?
 - **RQ4 (Edge Compression Viability):** Can ultra-compact student models preserve high linguistic fidelity after Knowledge Distillation (Knowledge Distillation (KD)) and INT8 quantization?
 - **RQ5 (Candidate Generation Utility):** Does a multi-candidate (Top- k) generation and reranking mechanism enhance user-perceived utility in interactive typing scenarios?
 - **RQ6 (Privacy and Adaptation):** Can lightweight, on-device personalization mechanisms adapt to user-specific abbreviations while upholding strict privacy constraints?

5. Project Objectives

To systematically address the research questions, the project defines the following specific, measurable, and achievable objectives:

1. **Dataset and Taxonomy Formalization:** Construct the **JDWR v1** multi-domain dataset (> 840,000 sentences across 8 distinct news domains) and formalize a realistic 6-dimensional typing noise taxonomy modeling virtual **QWERTY!** keyboard geometry.
2. **Open Architectural Benchmarking:** Formulate an open model-selection protocol to systematically evaluate candidate model families (including recurrent Bidirectional Gated Recurrent Unit (BiGRU)/Bidirectional Long Short-Term Memory (BiLSTM), convolutional Temporal Convolutional Network (TCN), and attention-based architectures) under identical data splits and evaluation constraints.
3. **Multi-Task Conditioning Exploration:** Investigate hierarchical multi-task neural architectures capable of jointly predicting character corrections, word segmentation boundaries, and diacritic tags without inter-task error propagation.
4. **Edge-First Optimization Framework:** Establish model compression and distillation pipelines to reduce memory footprint by > 70% and ensure sub-millisecond per-sentence inference on consumer-grade mobile CPU hardware.

-
5. **Multi-Tier Evaluation and Delivery:** Implement a standardized 3-tier evaluation harness (spanning Character, Word, and Sentence-level metrics) and deliver reproducible demonstration interfaces (Representational State Transfer (REST) Application Programming Interface (API) and interactive web User Interface (UI)).

6. Significance of the Project

- **Theoretical and Scientific Significance:** NextKey advances the state of Vietnamese NLP by shifting the academic paradigm from narrow, isolated diacritic restoration to realistic, compound-noise compact writing restoration. It provides empirical insights into architectural scaling behaviors, topological trade-offs, and multi-task feature gating under character-level language constraints.
- **Practical and Technological Value:** The resulting ultra-compact models offer an immediately deployable, zero-latency inference engine suitable for integration into embedded mobile operating systems, virtual keyboards, and assistive accessibility tools.
- **User Privacy and Societal Impact:** By enabling complete on-device execution without transmitting keystroke streams to remote cloud servers, NextKey provides absolute data sovereignty and privacy protection for millions of Vietnamese digital device users.

7. Project Scope & Limitations

7.1. In-Scope

- Multi-domain Vietnamese linguistic data covering journalism, administration, everyday conversation, and general texts.
- Joint modeling of missing diacritics, word boundary concatenation, and spatial keyboard-adjacent typographical errors.
- Open empirical exploration across multiple neural sequence modeling families and compression techniques (KD, Post-Training Quantization (PTQ), Quantization-Aware Knowledge Distillation (QKD)).
- Standalone evaluation harnesses, modular REST API endpoints, and web-based interactive demonstration prototypes.

7.2. Out-of-Scope & Limitations

- Acoustic speech-to-text processing or multimodal input modalities.

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- Low-level kernel driver development for specific proprietary mobile operating systems (the project provides standard SDK/API interfaces for core inference).
 - Highly specialized ancient historical scripts or rare ethnic minority dialectal forms beyond standard contemporary Vietnamese.

II. Project Management Plan

1. Team Work

1.1. Team Structure and Roles

The NextKey project is conducted by a dedicated team of three Artificial Intelligence engineering undergraduates at FPT University. To balance theoretical exploration with robust software engineering, responsibilities are organized into specialized research and development tracks:

Role	Assignee	Core Responsibilities
Lead AI Researcher & Architecture Track	Team Member 1	• Formulate mathematical problem objectives and loss functions. • Design sequence modeling architectures and multi-task feature flows. • Oversee baseline benchmarking and model-selection decision gates. • Lead academic thesis drafting and theoretical error analysis.
Edge Systems & Model Optimization Track	Team Member 2	• Formulate Knowledge Distillation (KD) loss formulations. • Implement PTQ and QKD quantization pipelines. • Benchmark CPU latency, throughput, and memory consumption. • Manage distributed training runs and GPU resource allocation.
Data Engineering & Software Track	Team Member 3	• Curate the multi-domain JDWR v1 dataset (847K sentences). • Develop the 6-dimensional keyboard-adjacency synthetic noise generator. • Implement FastAPI backend services and interactive Streamlit UI demo. • Maintain WBS schedule tracking and automated testing pipelines.

Table II.1: Team Structure, Research Tracks, and Role Allocations

1.2. Communication Plan

To ensure methodological alignment, rapid issue resolution, and steady milestone progression, the team adheres to a multi-tiered communication framework:

- **Daily Standups (Asynchronous / 15 Minutes):** Conducted via dedicated Slack/Discord channels to track daily commitments, synchronize code changes, and identify technical blockers early.
- **Weekly Advisor Consultation (1 Hour):** Formal weekly meetings with the Academic Supervisor to review experimental logs, evaluate empirical trade-offs, refine theoretical formulations, and validate subsequent sprint objectives.
- **Version Control and Collaborative Engineering:** Centralized GitHub repository with branch protection rules. All code additions require structured pull requests, passing automated test suites, and peer review prior to integration.
- **Shared Research Registry:** Systematized tracking of model configurations, random seeds, training checkpoints, evaluation logs, and hyperparameter matrices across all experimental phases.

2. Project Management Approach

2.1. Work Breakdown Structure

The research and development lifecycle is organized into six structured Work Packages (WPs) spanning a 16-week timeline, following an empirical and iterative milestone-driven methodology:

WBS ID	Work Package / Milestone	Scope & Key Deliverables	Timeline	Owner
WP 1.0	Research Scoping & Foundation	Problem definition, research contract, taxonomy design, repository scaffold	Weeks 1–2	All
WP 2.0	Data Pipeline & Noise Engine	JDWR v1 dataset curation, spatial QWERTY noise generator, validation protocol	Weeks 3–4	M3
WP 3.0	Baseline & Backbone Search	Implementation of rule heuristics, BiGRU, LSTM, CNN, TCN, and Transformer	Weeks 5–6	M1
WP 4.0	Topological Scaling & Search	Width vs depth exploration, 1-layer wide paradigms, ultra-small student sizing	Weeks 7–8	M1, M2
WP 5.0	Edge Distillation & Quantization	Knowledge Distillation, INT8 PTQ/QKD pipelines, CPU latency profiling	Weeks 9–10	M2
WP 6.0	Multi-Task Conditioning	Joint correction, boundary, and diacritic gating architectures	Weeks 11–12	M1, M2
WP 7.0	System Integration & Demo	FastAPI REST backend, Streamlit interactive UI, client SDK prototype	Weeks 13–14	M3
WP 8.0	Evaluation & Final Thesis	Cross-domain benchmark analysis, error taxonomy, final thesis, and defense slides	Weeks 15–16	All

Table II.2: Work Breakdown Structure (WBS) and Research Project Timeline

2.2. Risk Management

A proactive Risk Register is established to identify potential methodological, operational, and computational challenges, along with concrete mitigation strategies:

Identified Risk	Sev.	Prob.	Potential Research Impact	Proactive Mitigation Strategy
Scope Creep	High	Med	Delayed completion of core objectives due to over-extended feature exploration	Strictly enforce P0/P1 prioritization; defer complex user-specific adapter training to future work.
Synthetic-to-Real Domain Gap	High	Med	High accuracy on synthetic test sets failing to generalize to authentic human typing	Benchmark against isolated external OOD domain (<i>the_thao</i>); calibrate noise probabilities using real user error patterns.
Quantization Degradation	Med	Low	Severe drop in linguistic restoration fidelity after INT8 weight quantization	Integrate Knowledge Distillation prior to PTQ; establish strict CER parity thresholds between FP32 and INT8.
Compute Resource Constraints	High	Med	Training queue bottlenecks during large-scale corpus exploration	Implement memory-efficient data loaders; leverage distributed multi-GPU Kaggle cloud allocations.
Keystroke Privacy Concerns	High	Low	User hesitation and privacy risks associated with logging private input	Guarantee 100% client-side, offline edge inference; enforce zero remote data transmission.

Table II..3: Project Risk Register and Mitigation Framework

2.3. Quality Management Plan

To ensure academic rigor, empirical reproducibility, and software reliability:

- **Strict Reproducibility Protocols:** All dataset generation splits, random shuffles, weight initializations, and training runs are seeded deterministically with seed 42.
- **Automated Testing Suite:** Continuous testing via `pytest` verifying data transformation schemas, tensor dimensionality alignment, loss backward flows, and inference pipelines.
- **Standardized 3-Tier Metric Harness:** Candidate models are universally evaluated across Character-level (CER, Boundary F1-Score (BF1)), Word-level (Word Error Rate (WER), Word Accuracy (WAcc)), and Sentence-level (Exact Match (EM), Near-Perfect $\leq 5\%$, Bilingual Evaluation Understudy (BLEU)-4, Recall-Oriented Understudy for Gisting Evaluation (ROUGE)-L) metrics.
- **Zero Test Contamination:** The 159,172 external *the_thao (Sports)* sentences remain completely isolated from training and hyperparameter tuning to ensure uncompromised out-of-distribution evaluation.

2.4. Budget and Resource Allocation

The NextKey project operates under an open-source, resource-efficient model with zero capital expenditure:

-
- **Computational Infrastructure:** Cloud-hosted dual Tesla T4 GPUs (via Kaggle and Google Colab academic allocations) combined with local Apple Silicon workstations for rapid prototyping.
 - **Software Stack:** Fully open-source ecosystem including PyTorch, Hugging Face Transformers, FastAPI, Streamlit, and Git.
 - **Total Direct Financial Budget:** \$0.00 USD.

2.5. Change Management Process

To accommodate necessary technical adaptations without derailing the overall project schedule:

1. **Gate 1 (Smoke Run Gate):** Any candidate neural architecture that fails to demonstrate training loss convergence within 1,000 steps or exhibits unacceptable latency is discarded early.
2. **Gate 2 (Objective Benchmark Gate):** Final architectural decisions must be strictly justified by quantitative metric improvements on shared validation splits rather than subjective preference.
3. **Kill Criteria for Extensions:** Secondary research extensions (e.g., dynamic online personalization) are postponed immediately if the primary multi-task restoration pipeline requires further stabilization.

2.6. Project Closure and Deliverables

Upon reaching the final project milestone:

- **Artifact and Checkpoint Handover:** All quantized checkpoints, configuration manifests, vocabulary mappings, and evaluation JSON reports are archived with cryptographic SHA-256 checksums.
- **Academic Documentation:** Complete, reproducible source code repositories, benchmark logs, and technical defense slides prepared for capstone examination.

III. Methodology

[Provide the final methodology following the template as part II in the Report #4]

[This section will detail how you will approach the project, with a focus on applying and implementing AI techniques. The team is allowed to customize the content of this section, however, the team needs to provide a detailed view of the methods and techniques used. To ensure that the reader understands the scientific and systematic approach that you apply to your capstone project in the field of AI. Below are two sample templates for this section that you can choose from or customize.]

1. Overview of Framework/Approach

[Describe the whole system in general.]

2. Data Collection and Preprocessing

[Describe how you will collect data, including the sources of data, methods of collection, and any selection criteria. Information about the sample size, diversity, and representativeness of the data should be included.]

[Detail how the data will be processed before analysis, including cleaning data, handling missing data, transforming data, and normalization.]

3. Feature Selection and Engineering

[Describe the process of selecting and creating features that will be used for the machine learning model, as well as the reasons for these choices.]

4. Proposed Model/System/Method (Contributions)

[Describe the contributions of the capstone project in detail.]

5. Experimental Settings

[Describe the machine learning models considered and the reasons for the final model choice, including technical specifications and configurations of the model.]

[Detail how the model will be trained and validated, including information on data splitting methods, cross-validation techniques, and the metrics used for evaluation.]

6. Evaluation Metrics

[Identify the metrics that you will use to assess the performance of the model, such as accuracy, F1 score, ROC AUC, etc.]

IV. Results and Discussion

[Provide the final results and discussion following the template as part II in the Report #6]

1. Results and Analysis

[Present the results obtained from executing the project, including numerical data, graphs, diagrams, and supportive imagery. This may include the outcomes of machine learning models, such as accuracy, confusion matrices, ROC curves, and other evaluation metrics.]

[Provide a detailed analysis of the results obtained, explaining their significance and impact on the project. This might include comparing the performance of different models, error analysis, and interpreting the features of the model.]

2. Discussion

[Discuss the results, including how they address the research questions and hypotheses, their alignment with results from related studies, and the broader context of the project. Mention any limitations or challenges encountered during the project.]

3. Personal Reflection (Optional)

[A section reflecting personally on the project process, lessons learned, and how improvements could be made in the future.]

[Note: The "Results and Discussion" section is an essential part of the capstone project document, where you evaluate the success of your project based on data and analysis. This is also an opportunity for you to reflect and discuss the larger significance of the work you have done, as well as to identify any opportunities for further research or development.]

V. System Design and Implementation

[Provide the final system design and implementation following the template as part II in the Report #5]

1. AI Model Integration

[Describe how the AI model is integrated into the larger system, including the interface and data exchange between the AI model and other system components.]

2. Data Flow and Processing

[Illustrate and describe the data flow through the system, from data collection, processing, to data analysis and result presentation.]

3. Deployment Strategy

[Detail how the system will be deployed, including the deployment environment and deployment procedures.]

4. Scalability and Maintenance

[Discuss how the system is designed to handle growth and future maintenance, including system updates, scalability, and version management.]

[Note: By providing a detailed view of the AI solution's design and implementation in this section, you will help readers understand the technical and systematic approach you have applied to your capstone project.]

VI. Conclusion

[Provide the final conclusion following the template as part II in the Report #7]

1. Summary of Findings

[Summarize the main findings of the project, including how the research questions have been answered and how the project goals were achieved.]

2. Contributions and Reflections on the Project

[Highlight the specific contributions of the project to the field of study, community, or industry. This could include technical advancements, new knowledge, or practical applications.] [Reflect on the project process, including what worked well and what could be improved. This is also an opportunity to acknowledge the level of challenge and learning from the project.] [Provide some final thoughts or conclusion, expressing hopes or vision for the future of the research or the problem addressed by the project.]

3. Limitations and Future Work

[Acknowledge any limitations of the project and how they may affect the results and conclusions. This includes factors such as data, methods, or assumptions.] [Suggest directions for future research or development based on the results and limitations of the project. This helps guide others who may build upon your work.]

[Note: The "Conclusion and Future Work" section is not only a place to summarize the completed work but also an opportunity to highlight the importance of the project and propose next steps. This helps readers understand the impact and potential of the work you have done.]

VII. Ethical Declaration

1. Ethical Considerations

[Discuss ethical issues that may arise when applying machine learning, including data privacy, accuracy, and how to handle data fairly.]

2. Declaration of AI-based Assistance

[Declare AI-based tools that have been used during doing the capstone project in detail.]

VIII. Appendices

This section can be removed out of the report if the capstone project does not contain any appendices.

1. First Appendix Sample

[Provide the final appendices following the template as part II in the Report #9]

[If there is any supplementary information that does not fit into the main sections of the document but is still important and useful for a full understanding of the project, you can include it in the appendices. This might include raw data, source code, full versions of surveys, more detailed diagrams, or other supporting documents for your main content.]

[Note: After including all these sections, ensure that all parts of the document are logically and coherently organized, and that you have thoroughly checked for language, spelling, and grammar errors.]

2. Second Appendix Sample

[Note: After including all these sections, ensure that all parts of the document are logically and coherently organized, and that you have thoroughly checked for language, spelling, and grammar errors.]

This part presents the instructions of the format of the references.

To manage the references, some tools, such as mendeley or jabref, have been recommended because the reference will only appear in referecne list once it is cited in the report. Try it!

You have to cite at least 1 paper, such as LSTM [4], Fast R-CNN [?], Transformer [3] or EfficientDet [?], if IEEE style \bibliographystyleIEEEtran and its bibliography style \bibliographyreferences have been used for managing and citing references.

[Provide the final references following the template as part II in the Report #8]

[Formatting: Ensure that all references are listed in a uniform and correct format, following the guidelines of the citation system you have chosen (e.g., APA, MLA, Chicago, etc.). This helps ensure accuracy and ease of source verification.]

[Order: Typically, reference items are arranged in alphabetical order for easy lookup. Make sure that all sources are listed comprehensively and without omission.]

[Completeness: For each reference, provide enough information so that readers can find and review the source material. This includes the author's name, year of publication, title of the work, publisher name or journal name, and other relevant information such as page numbers or URL.]

[Accuracy: Carefully check to ensure that all information about the references is correct and up-to-date.] [Relevance: Only include sources that were actually cited or used in your project. This helps maintain the integrity and accuracy of the project.]

[Note: The "References" section is an indispensable part of any academic project. It not only proves that you have conducted careful and grounded research but also allows future generations of students to check and expand their projects from the sources you have cited.]

[You should remove this part out of the final manuscript.]

IX. References

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